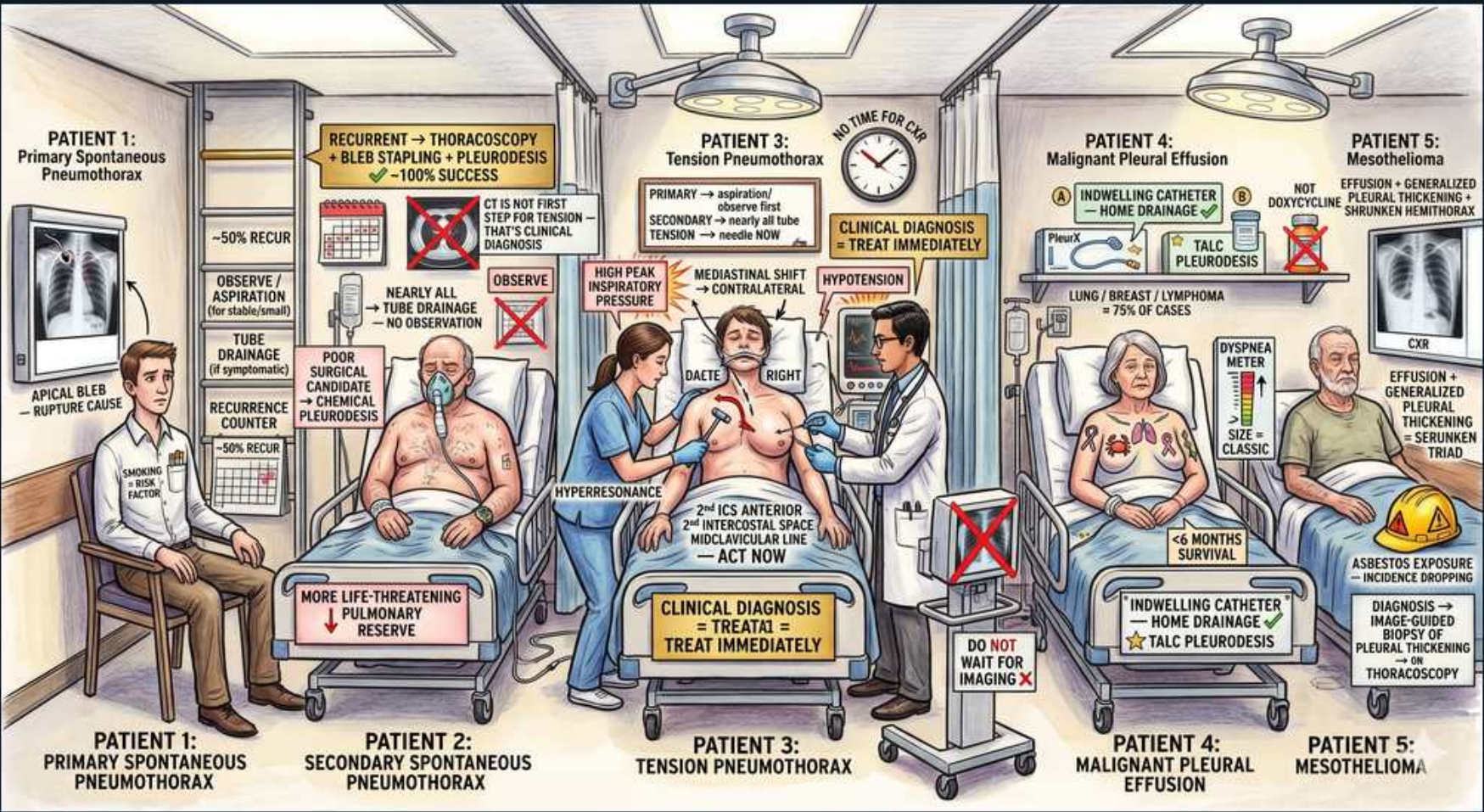


Pleural Disease — Pneumothorax & Malignant Effusion



1. Primary spontaneous PTX

WHAT YOU SEE

CXR with apical bleb; 'PATIENT 1: PRIMARY SPONTANEOUS PNEUMOTHORAX'

WHAT IT ENCODES

Primary spontaneous pneumothorax (PSP) occurs WITHOUT clinically apparent lung disease, classically in a tall, thin, young man (teens-30s) who smokes. Mechanism: rupture of an apical subpleural bleb/bulla. Smoking raises risk ~9-20x; there is also an association with Marfan and a positive family history.

BOARD TRAP

'Primary' does not mean idiopathic/risk-free — SMOKING is the key modifiable risk, and cessation reduces recurrence. The trap is treating PSP as purely random; counsel smoking cessation, which lowers the high recurrence rate.

2. Apical bleb rupture

WHAT YOU SEE

Label 'APICAL BLEB RUPTURE CAUSE'

WHAT IT ENCODES

The anatomic cause of PSP is rupture of an apical subpleural bleb, releasing air into the pleural space. Blebs form preferentially at the apex due to higher distending pressures there; they are often bilateral, which is why contralateral recurrence can occur.

BOARD TRAP

Don't assume a single bleb — blebs are frequently multiple and bilateral, so finding/treating one apex does not eliminate the risk on the other side. This is why recurrent or bilateral disease prompts surgical bleb resection + pleurodesis.

3. Small PSP: observe

WHAT YOU SEE

Box 'OBSERVE / ASPIRATION', '<50% recur'

WHAT IT ENCODES

Management of PSP is size- and symptom-based: small (< 2-3 cm at hilum or < ~20%) and minimally symptomatic → observation + supplemental high-flow O2 (O2 accelerates pleural air reabsorption ~4x by creating a nitrogen gradient). Larger or symptomatic PSP → needle aspiration or a small-bore catheter/chest tube.

BOARD TRAP

Not every PSP needs a chest tube — a small, stable, asymptomatic PSP is observed with supplemental O2. The trap is reflexively placing a chest tube in every pneumothorax; oversized intervention adds morbidity for a PSP that would resorb.

4. Recurrence counter

WHAT YOU SEE

Label 'RECURRENCE COUNTER', tube drainage

WHAT IT ENCODES

PSP recurrence is high (~30% after a first event, rising with each subsequent episode). A first recurrence (or persistent air leak > 3-5 days, bilateral PTX, or high-risk occupation such as pilot/diver) is an indication for definitive surgery with pleurodesis to prevent future events.

BOARD TRAP

Treating a recurrence the same as a first event (just another tube) misses the indication for definitive prevention — the trap is not offering VATS/pleurodesis after recurrence. Recurrent PSP warrants surgical management, not endless drainage cycles.

5. Secondary spontaneous PTX

WHAT YOU SEE

Older patient with mask; 'PATIENT 2: SECONDARY SPONTANEOUS PNEUMOTHORAX'

WHAT IT ENCODES

Secondary spontaneous pneumothorax (SSP) arises on a background of lung disease — COPD/emphysema is the most common, plus asthma, cystic fibrosis, Pneumocystis (PCP) and other necrotizing infections, lung cancer, Marfan, and LAM (young women). The diseased lung has little reserve, so SSP is more symptomatic and dangerous than PSP.

BOARD TRAP

SSP is more dangerous than PSP because of poor pulmonary reserve — admit and place a chest tube even when the pneumothorax is small. The trap is applying the 'observe small PTX' rule from PSP to SSP; in SSP, even a small one warrants drainage and admission.

6. More life-threatening

WHAT YOU SEE

Banner 'MORE LIFE-THREATENING — PULMONARY RESERVE'

WHAT IT ENCODES

Because underlying lung disease leaves minimal functional reserve, SSP is poorly tolerated and carries higher morbidity/mortality. Management is almost always chest tube drainage + hospital admission, with a lower threshold to proceed to surgery/pleurodesis given the high recurrence and poor tolerance.

BOARD TRAP

Do NOT simply observe an SSP — low reserve means even a small one can rapidly decompensate. The trap is conservative outpatient management; SSP needs admission and a chest tube.

7. Surgery / pleurodesis

WHAT YOU SEE

'SURGICAL CANDIDATE → CHEMICAL PLEURODESIS'

WHAT IT ENCODES

Definitive prevention is VATS with bleb/bulla stapling (resection) PLUS pleurodesis (mechanical abrasion or chemical, e.g., talc) to obliterate the pleural space. Indications: recurrent PTX, persistent air leak (> 3-5 days), bilateral PTX, first PTX in high-risk occupations, or first SSP in many practices.

BOARD TRAP

Chemical pleurodesis alone for an ongoing large air leak without addressing the bleb may fail — the trap is choosing pleurodesis when the lung won't stay up. VATS resection of the culprit bleb combined with pleurodesis is the durable answer for recurrence/persistent leak.

8. Tension pneumothorax

WHAT YOU SEE

Patient in distress, trachea pushed; 'PATIENT 3: TENSION PNEUMOTHORAX'

WHAT IT ENCODES

Tension pneumothorax: a one-way valve lets air enter the pleural space on inspiration but not escape, progressively raising intrapleural pressure → mediastinal shift, kinking of the great veins, and obstructive shock. Signs: severe hypotension, distended neck veins (impaired venous return), tracheal deviation AWAY from the affected side, absent breath sounds, and hyperresonance on the affected side.

BOARD TRAP

Tension PTX is a CLINICAL diagnosis — do NOT wait for a chest X-ray. The trap is ordering imaging in an unstable patient; treat immediately with decompression. Note the JVD here is from obstructed venous return, paralleling tamponade/RV infarct as a cause of hypotension + JVD + clear-ish auscultation.

9. Clinical Dx — act now

WHAT YOU SEE

Banner 'CLINICAL DIAGNOSIS = TREAT IMMEDIATELY'

WHAT IT ENCODES

Recognize tension physiology at the bedside: hypotension, tachycardia, hyperresonance and absent breath sounds on the affected side, contralateral tracheal shift, JVD, and (in a ventilated patient) suddenly rising peak inspiratory pressures with desaturation. Immediate decompression is both diagnostic and therapeutic.

BOARD TRAP

'Do NOT wait for imaging' — ordering a CXR before decompression is the classic wrong board answer for an unstable tension PTX. The trap is prioritizing confirmation over treatment; in the unstable patient, needle/finger decompression comes first.

10. Needle decompression site

WHAT YOU SEE

'2nd ICS ANTERIOR MIDCLAVICULAR LINE — ACT NOW'

WHAT IT ENCODES

Immediate needle (or finger) thoracostomy: insert a large-bore (14-16 G) needle in the 2nd intercostal space at the midclavicular line, just above the 3rd rib — OR at the 5th intercostal space in the anterior axillary line (often preferred in adults/larger chests for better success). This converts a tension PTX to a simple one; a chest tube must then follow.

BOARD TRAP

Needle decompression is only the IMMEDIATE temporizing step — the definitive treatment is a chest tube afterward. The trap is stopping at the needle and not placing a tube, or choosing the wrong site/below the rib (risking the neurovascular bundle). Always: decompress, then tube.

11. Mediastinal shift

WHAT YOU SEE

Label 'MEDIASTINAL SHIFT → CONTRALATERAL', high peak inspiratory pressure

WHAT IT ENCODES

In tension PTX the trachea and mediastinum shift TOWARD the unaffected (contralateral) side as pressure builds, compressing the opposite lung and the great veins. In a mechanically ventilated patient, an abrupt rise in peak inspiratory/airway pressures with hypotension and desaturation is an early, easily missed clue.

BOARD TRAP

Contrast directions: tension PTX pushes the trachea AWAY (contralateral); a large atelectasis/lobar collapse or fibrosis PULLS the trachea TOWARD the affected side. The trap is reversing the direction — deviation toward the affected side is collapse, not tension.

12. Malignant pleural effusion

WHAT YOU SEE

Woman in bed; 'PATIENT 4: MALIGNANT PLEURAL EFFUSION', indwelling catheter

WHAT IT ENCODES

Malignant pleural effusion: lung, breast, and lymphoma cause ~75%; the fluid is an exudate, frequently large, recurrent, and bloody, with low glucose/pH in high tumor burden. Its presence signifies metastatic/advanced disease with a median survival often < 6 months, so management is palliative and goal-directed.

BOARD TRAP

For a RECURRENT malignant effusion, choose an indwelling tunneled pleural catheter or talc pleurodesis — repeated therapeutic thoracenteses are not the long-term answer. The trap is tapping the patient over and over instead of providing a definitive palliative drainage strategy.

13. Talc pleurodesis

WHAT YOU SEE

Labels 'TALC PLEURODESIS', 'INDWELLING CATHETER → HOME DRAINAGE'

WHAT IT ENCODES

Definitive palliation of recurrent malignant effusion: talc pleurodesis (instilled talc induces inflammation that fuses visceral and parietal pleura — REQUIRES an expandable lung) OR a tunneled indwelling pleural catheter (IPC) for intermittent home drainage (works even with trapped lung and can produce spontaneous pleurodesis over time).

BOARD TRAP

Pleurodesis FAILS if the lung is trapped/non-expandable (it cannot appose the parietal pleura) — use an indwelling catheter instead. The trap is attempting talc pleurodesis in a trapped lung; confirm lung re-expansion after drainage before committing to pleurodesis.

14. Mesothelioma

WHAT YOU SEE

Older man; 'PATIENT 5: MESOTHELIOMA', asbestos, CXR pleural thickening

WHAT IT ENCODES

Malignant pleural mesothelioma: strong association with ASBESTOS exposure and a long latency (20-40 years); presents with a unilateral effusion plus nodular/circumferential pleural thickening and chest pain. Smoking does NOT cause mesothelioma but multiplies asbestos-related bronchogenic carcinoma risk. Diagnosis requires tissue.

BOARD TRAP

Fluid cytology is UNRELIABLE for mesothelioma — get a thorascopic (or image-guided core) pleural BIOPSY, not just fluid cytology. The trap is excluding mesothelioma based on negative cytology; the diagnosis (and distinguishing it from adenocarcinoma via immunostains) needs adequate tissue.